A Pre-Registered Direction Instability Atlas Across Three Perturbation Modalities

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Abstract

Background: When a drug or gene knockout is applied across many cell lines, the resulting expression or morphological signatures can point in different directions—a property we term direction instability (DI). Because DI normalizes each signature to unit length before comparison, it is immune to the first-order magnitude confound that dominates alternative geometric metrics. A companion study established this immunity and showed that DI predicts cross-cell-line mechanism transport (pre-registered, construction-neutral AUROC = 0.945).

Results: We constructed a cross-modal DI atlas spanning three measurement technologies—LINCS L1000 transcriptomics (8,949 drugs), Tahoe-100M single-cell transcriptomics (379 drugs), and JUMP Cell Painting morphology (7,948 CRISPR knockouts, 12,590 ORF overexpressions)—providing magnitude-corrected DI with bootstrap confidence intervals for every perturbation. Using eight pre-registered primary tests and one exploratory analysis, we characterized what DI does and does not capture. DI is concordant across expression and morphology modalities (partial $\rho = 0.19$, $n = 378$ compounds, $p = 2.1 \times 10^{-4}$), confirming that it reflects biology rather than measurement artifact. DI cannot, however, be reduced to simpler gene-level features. Essential genes show *lower* DI ($\rho = -0.33$), inverting the pre-registered prediction: functional constraint produces stereotyped disruption, suppressing context-dependence. Knockout and overexpression DI for the same gene are uncorrelated ($\rho = 0.01$, $n = 5{,}220$), establishing DI as a property of the perturbation mechanism rather than the target gene. Expression variance, paralog count, and paralog expression breadth each explain under 0.4% of DI variance after confound control.

Conclusions: DI captures a cross-modally reproducible interaction between perturbation mechanism and cellular context that existing gene annotations do not. The atlas, pre-registration documents, and analysis code are publicly available.

1 Introduction

High-throughput perturbation screens now profile thousands of drugs and genetic perturbations across panels of cell lines, producing multi-dimensional signatures in transcriptomic (Subramanian et al., 2017), morphological (Bray et al., 2016), and single-cell modalities (Tahoe Consortium, 2025). A central question in perturbation biology is whether a compound or knockout acts the same way across cellular contexts or produces context-dependent effects. This question has practical consequences: drugs with context-invariant mechanisms are more likely to transfer from the cell line where they were discovered to the patient tissue where they must work (Tower, 2026).

Direction instability (DI) quantifies this context-dependence as $\mathrm{DI} = 1 - \overline{\cos(\mathbf{s}_i, \mathbf{s}_j)}$, the mean pairwise cosine dissimilarity among a perturbation's signatures across contexts. Because cosine normalizes each signature to unit length before comparison, DI is immune to the first-order magnitude confound—systematic covariation between effect size and metric value—that dominates alternative geometric measures such as Grassmannian geodesic distance, profile variability, and optimal transport cost (Tower, 2026). A residual second-order correlation persists near zero (noise-dominated signatures inflate pairwise cosine dissimilarity), which we remove by magnitude correction (Section 2). A companion study demonstrated this immunity across three perturbation modalities and showed that DI predicts cross-cell-line mechanism transport (pre-registered, construction-neutral AUROC = 0.945 on 8,949 LINCS drugs; Tower, 2026). The present study takes DI's validity as established and asks a different question: what does DI capture, and what does it fail to capture, across modalities and perturbation types?

We constructed a DI atlas spanning three measurement technologies and four perturbation classes: drug perturbations measured by L1000 transcriptomics (LINCS, 8,949 compounds) and single-cell transcriptomics (Tahoe-100M, 379 compounds); genetic knockouts measured by Cell Painting morphology (JUMP-CP CRISPR, 7,948 genes); and genetic overexpressions measured by Cell Painting morphology (JUMP-CP ORF, 12,590 genes). For each perturbation, the atlas provides magnitude-corrected DI with bootstrap confidence intervals, enabling downstream analyses that require a calibrated measure of context-dependence.

To characterize this resource, we pre-registered eight primary hypotheses across two batches before computing DI on new datasets (Batch 1 SHA `0d07a01`; Batch 2A SHA `a17c125`). The hypotheses test whether DI is concordant across modalities, whether it correlates with gene essentiality, whether it is an intrinsic property of genes, and whether expression variance, drug selectivity, or paralog buffering can explain it. Of these, one passes its pre-registered criterion and seven fail. A ninth test—PRISM viability concordance—is exploratory.

The failures are the core result. Cross-modal concordance confirms that DI reflects biology rather than measurement artifact: compounds whose transcriptomic effects vary across cell lines also produce variable morphological profiles. Every other test delimits what DI measures. Essential genes show *low* DI, because functional constraint produces stereotyped disruption regardless of cellular identity. Knockout and overexpression DI are unrelated, meaning DI characterizes how a specific perturbation mechanism interacts with context rather than indexing a stable property of the target gene. Expression variance, paralog count, and paralog expression breadth produce correlations too small to matter biologically (partial $\rho < 0.07$ after confound control), ruling them out as explanations for DI variation.

The failure pattern itself is the finding: DI measures something that existing gene annotations do not, and the atlas makes that measurement available at scale.

The remainder of the paper is organized as follows. Section 2 describes the atlas as a resource: datasets, DI computation, and magnitude correction. Section 3 presents the construct validity results, with PRISM viability as independent corroboration. Section 4 reports the informative null and sign-flipped results that delimit what DI measures. Section 5 describes the pre-registration protocol and deviations. Section 6 synthesizes the failure pattern into a characterization of DI as an irreducible perturbation property.

2 The DI atlas as a resource

2.1 Datasets

The atlas draws on five perturbation datasets spanning three measurement technologies (Table 1). Two drug datasets were previously analyzed in the companion confound study (Tower, 2026): LINCS L1000 transcriptomics (978 landmark genes, 8,949 compounds across variable numbers of cell lines; Subramanian et al., 2017) and JUMP-CP Cell Painting compound morphology (3,180 features, 25,254 compounds; Chandrasekaran et al., 2024). Three datasets are new to this study: Tahoe-100M single-cell transcriptomics (379 drugs across 47–50 cell lines per drug; Tahoe Consortium, 2025), JUMP-CP CRISPR knockouts (7,948 genes, 259 post-sphering PCA features, plate replicates as contexts; Chandrasekaran et al., 2025), and JUMP-CP ORF overexpressions (12,590 genes, 722 features, plate replicates as contexts; Chandrasekaran et al., 2025).

Table 1: Datasets in the DI atlas. "Contexts" denotes the unit over which DI is computed: cell lines for drug datasets, plate replicates for genetic perturbation datasets. Only perturbations with ≥ 5 contexts are included. The pre-registration estimated 15,121 ORF genes; the actual count after quality filtering is 12,590.

Dataset	**Modality**	**Perturbations**	**Features**	**Contexts**	**Source**
LINCS L1000	Transcriptomic	8,949 drugs	978	Cell lines	Pre-computed
Tahoe-100M	Transcriptomic	379 drugs	∼62,700	Cell lines	HuggingFace
JUMP-CP compounds	Morphological	25,254 compounds	3,180	Source ctx.	Pre-computed
JUMP-CP CRISPR	Morphological	7,948 genes	259	Plate reps	CPG
JUMP-CP ORF	Morphological	12,590 genes	722	Plate reps	CPG

CPG = Cell Painting Gallery (cpg0016).

A sixth dataset, PRISM repurposing viability (Corsello et al., 2020), is included as a complementary validation source. PRISM measures scalar cell viability (log-fold-change) rather than high-dimensional profiles, so cosine-based DI is undefined. We compute the coefficient of variation (CV) of viability across cell lines as a scalar analogue of context-dependence. PRISM results are labeled exploratory throughout.

2.2 DI computation

For a perturbation with $K \geq 5$ context-specific signatures $\mathbf{s}_1, \ldots, \mathbf{s}_K \in \mathbb{R}^d$, direction instability is

$$\mathrm{DI} = 1 - \frac{2}{K(K-1)} \sum_{i<j} \frac{\mathbf{s}_i \cdot \mathbf{s}_j}{\|\mathbf{s}_i\| \, \|\mathbf{s}_j\|}. \tag{1}$$

DI $= 0$ when all signatures point in the same direction (perfectly consistent perturbation) and approaches 1 when signatures are orthogonal or anti-correlated (maximally context-dependent).

2.3 Magnitude immunity and correction

The first-order magnitude confound arises when a metric's expected value changes with effect size: stronger perturbations mechanically produce larger distances (or smaller ones, depending on the metric). Cosine similarity is ratio-scale—$\cos(\alpha\mathbf{s}_i, \alpha\mathbf{s}_j) = \cos(\mathbf{s}_i, \mathbf{s}_j)$ for any $\alpha > 0$—so DI is invariant to uniform rescaling of all signatures. This is the sense in which DI is "immune by construction": the dominant confound pathway (stronger effect → larger norm → larger distance) is algebraically cancelled. The companion study (Tower, 2026) verified this empirically across three modalities (LINCS, DepMap CRISPR, Perturb-seq), showing that alternative metrics exhibit Cohen's $d = 0.55$–2.65 magnitude confounding while DI shows none after correction.

A residual second-order correlation remains. Weakly-acting perturbations produce noisier signature estimates, and noise inflates the average pairwise cosine dissimilarity even after unit-length normalization. We remove this residual confound by OLS regression of raw DI on mean signature L_2 norm:

$$\mathrm{DI}_{\mathrm{corrected}} = \mathrm{DI}_{\mathrm{raw}} - \hat{f}(\overline{\|\mathbf{s}\|}) + \hat{\beta}_0, \tag{2}$$

where $\hat{f}$ is the OLS prediction and $\hat{\beta}_0$ is the intercept. This correction is applied per dataset. $\mathrm{DI}_{\mathrm{corrected}}$ is the primary metric throughout; $\mathrm{DI}_{\mathrm{raw}}$ is reported for transparency.

2.4 Bootstrap confidence intervals

For each perturbation, we resample the K contexts with replacement 1,000 times and recompute DI on each bootstrap sample. The 2.5th and 97.5th percentiles define the 95% bootstrap confidence interval. These intervals quantify the stability of each DI estimate with respect to the particular set of contexts observed.

2.5 Statistical framework

All hypothesis tests use partial Spearman correlation with confound control. The partial correlation is computed by Pearson-residualizing both variables against the covariate matrix (with intercept column), then computing Spearman's ρ on the residuals. This method avoids the bias of rank-then-residualize approaches, which produce spurious partial correlations of $\rho \approx 0.05$–0.25 under the null at large n (verified in simulation; see pre-registration documents). Confidence intervals use the Fisher z-transform with degrees of freedom adjusted for the number of covariates.

Every primary test applies two decision criteria simultaneously: (i) $p < \alpha$ (Bonferroni-corrected: $\alpha = 0.0125$ for Batch 1, $\alpha = 0.0167$ for Batch 2A), and (ii) the 95% CI lower bound exceeds ± 0.05 in the predicted direction. The CI floor is necessary because at $n > 5{,}000$, effects as small as $\rho = 0.03$ reach statistical significance. Without the floor, trivially small correlations would pass.

2.6 Data format

The atlas is distributed as CSV files containing, for each perturbation: $\mathrm{DI}_{\mathrm{raw}}$, $\mathrm{DI}_{\mathrm{corrected}}$, bootstrap CI bounds, magnitude CV, mean signature norm, and number of contexts. Full data provenance and repository locations are listed in the Data Availability section.

3 Construct validity: cross-modal concordance

The atlas is useful only if DI reflects a biological property of perturbations rather than a measurement artifact of any single platform. Two pre-registered concordance tests assess this by comparing DI values for the same compounds measured in different datasets. An exploratory analysis using PRISM viability provides independent corroboration from a scalar outcome.

3.1 Cross-modal concordance (LINCS ∩ JUMP-CP)

Compounds present in both LINCS L1000 (transcriptomic DI) and JUMP-CP Cell Painting (morphological DI) were matched by InChIKey connectivity prefix (14 characters). Of the matched set, $n = 378$ compounds have ≥ 5 contexts in both datasets.

Partial Spearman $\rho = 0.190$ (95% CI $[0.090, 0.285]$, $p = 2.1 \times 10^{-4}$), controlling for LINCS n_{contexts} as one covariate. The CI lower bound (0.090) exceeds the 0.05 floor, and $p < 0.0125$. This test passes both pre-registered criteria.

The effect size falls in the "weakly concordant" bin ($0.15 \le \rho < 0.30$) defined in the pre-registration, consistent with the expectation that cross-modal comparisons capture only the fraction of biology shared between gene expression and cell morphology (Haghighi et al., 2022). The concordance is weak but genuine: compounds whose transcriptomic signatures vary across cell lines also produce variable morphological profiles, and this relationship survives confound control.

3.2 Same-modality concordance (LINCS ∩ Tahoe)

Drugs present in both LINCS L1000 and Tahoe-100M were matched by InChIKey connectivity prefix, verified via PubChem CID cross-reference. This yielded $n = 76$ overlapping compounds (fewer than the pre-registered estimate of $n \approx 103$, because some Tahoe drugs lack PubChem CID annotations mapping to LINCS InChIKeys). LINCS DI is computed in $\mathbb{R}^{978}$ (landmark genes) while Tahoe DI is computed in $\mathbb{R}^{\sim 62{,}000}$ (full transcriptome). Because cosine similarity is dimensionless and scale-invariant, DI values are comparable across feature spaces of different dimensionality; a supplementary analysis restricting Tahoe signatures to the 978 landmark genes would provide a direct dimensionality-matched comparison.

Partial Spearman $\rho = 0.254$ (95% CI $[0.027, 0.456]$, $p = 0.029$), controlling for LINCS and Tahoe n_{contexts} (two covariates). This test **fails**: the CI lower bound (0.027) does not clear the 0.05 floor, and $p = 0.029$ exceeds the Bonferroni threshold ($\alpha = 0.0125$).

The failure is one of statistical power, not absence of concordance. The point estimate ($\rho = 0.254$) is larger than the cross-modal result ($\rho = 0.190$), and the pre-registered power analysis predicted this outcome: at $n = 76$, the study achieves 80% power only for true effects ≥ 0.35. We interpret this pattern as a sample-size artifact—the LINCS-Tahoe overlap ($n = 76$) is far smaller than LINCS-JUMP ($n = 378$)—rather than evidence that DI transfers better across modalities than within one. A deviation in the matching method is disclosed in Section 5.

3.3 PRISM viability (exploratory)

As an independent check on construct validity, we computed the Spearman correlation between LINCS DI and PRISM viability CV for $n = 75$ shared drugs (filtered to $|\text{mean LFC}| \ge 0.01$). Spearman $\rho = 0.359$ (95% CI $[0.144, 0.542]$, $p = 0.0016$).

This result is **exploratory**: n is small, the p-value is uncorrected, and no covariate adjustment was applied. It should carry weight only alongside the primary tests. That said, viability is a scalar outcome with a fundamentally different noise structure from high-dimensional expression or morphology profiles. The correlation suggests that compounds whose transcriptomic effects depend on cellular context also show variable cell-killing potency, consistent with the interpretation that DI captures a biological property of drug-context interactions.

4 What DI is not: informative nulls and the essentiality sign-flip

Five pre-registered experiments tested whether DI variation across genes and drugs can be explained by simpler biological properties. All five fail their pre-registered criteria, and each failure constrains the interpretation of DI in a specific way. Table 2 summarizes the full scorecard.

4.1 Functional constraint suppresses DI

The pre-registration predicted that broadly essential genes would show high morphological DI: genes required in many cell types should produce context-dependent phenotypic disruption. The

Table 2: Pre-registered experiment results. All tests use partial Spearman ρ with confound control except the knockout vs. overexpression test (raw Spearman + permutation null). "CI gate" indicates that the 95% CI lower bound did not clear the ± 0.05 threshold in the predicted direction.

Hypothesis	n	**Partial** ρ	p	**Verdict**
Cross-modal concordance	378	0.190	2.1×10^{-4}	**PASS**
Same-modality concordance	76	0.254	0.029	FAIL (CI gate)
Essentiality $\rightarrow$ high DI	7,653	-0.332	~ 0	FAIL (wrong sign)
KO $\approx$ OE DI	5,220	0.011	0.43	FAIL (null)
Expression variance	7,802	0.060	1.1×10^{-7}	FAIL (CI gate)
Drug selectivity	75	0.144	0.22	FAIL (underpowered)
Paralog count	6,894	-0.007	0.57	FAIL (null)
Paralog breadth	6,882	-0.029	0.015	FAIL (CI gate)
Viability concordance	75	0.359	0.0016	Exploratory PASS

observed correlation is large, precisely estimated, and in the opposite direction. Partial Spearman $\rho = -0.332$ (95% CI $[-0.352, -0.312]$, $p \approx 0$, $n = 7{,}653$), controlling for n_{contexts}. Essentiality breadth (fraction of DepMap 24Q4 cell lines with Chronos score < -0.5; Funk et al., 2022) predicts *lower* DI, accounting for approximately 11% of variance.

The sign-flip has a straightforward interpretation, which we emphasize is *post-hoc*: the pre-registered prediction confused two senses of context-dependence. Essential genes affect many cell types, but they affect them in the same way. A gene required for mitosis, ribosome assembly, or DNA repair produces a stereotyped catastrophe regardless of cellular identity. Functional constraint makes the downstream phenotypic consequences uniform across contexts, yielding low DI. Genes with high DI tend to be selectively essential or non-essential—their knockout phenotypes interact with cell-type-specific pathways, producing variable consequences across the panel.

This interpretation generates a falsifiable prediction: among non-essential genes, those with tissue-restricted expression (active in few cell types) should show *higher* DI than ubiquitously expressed non-essential genes, because tissue-restricted function implies cell-type-specific downstream consequences. We flag this as a candidate for prospective pre-registration.

The result connects DI to established concepts in functional genomics: the phenotypic landscape of essential genes is structured by functional constraint (Funk et al., 2022; Ramezani et al., 2025), and DI indexes the complement of that constraint.

4.2 DI is mechanism-specific

If context-dependence were an intrinsic property of genes, then a gene's knockout DI and its overexpression DI should correlate. Spearman $\rho = 0.011$ (95% CI $[-0.016, 0.038]$, $p = 0.43$, $n = 5{,}220$ genes with ≥ 5 replicates in both CRISPR and ORF). A mandatory permutation null (1,000 label shuffles preserving plate structure; Appendix B) yielded a null distribution with mean = 0.0003, SD = 0.014, 95th percentile = 0.023. The observed $\rho = 0.011$ falls within the null (permutation $p = 0.215$).

At $n = 5{,}220$, the study had power to detect effects as small as $\rho = 0.04$ (SE ≈ 0.014). The null is well-powered: losing a protein and gaining excess protein engage different molecular

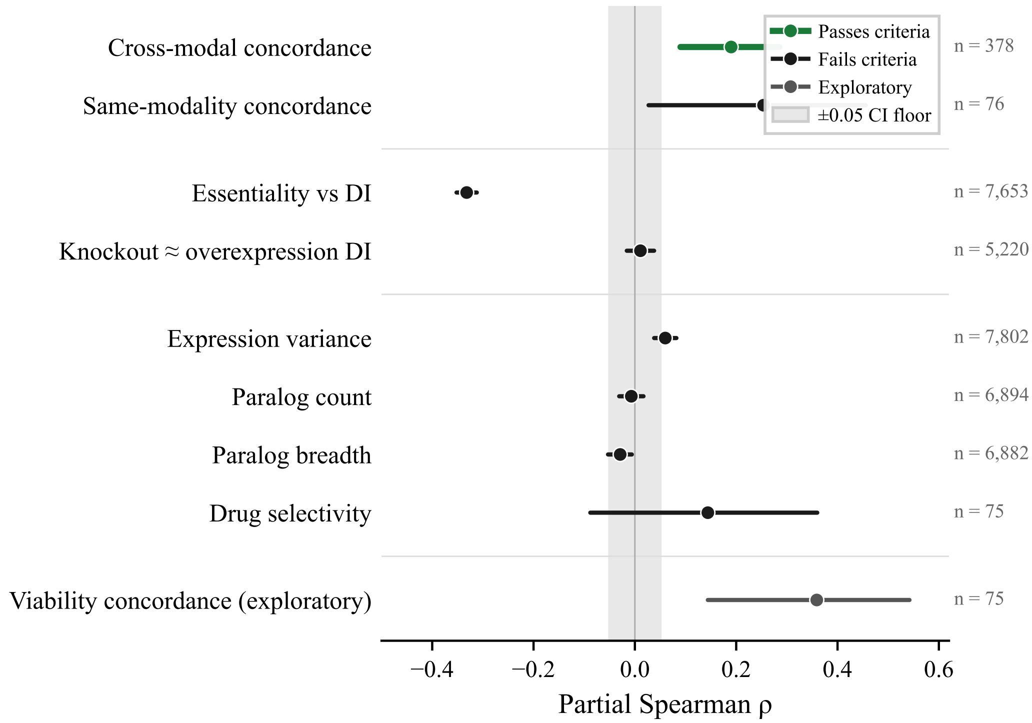

Figure 1: Pre-registered experiment results. Each point shows the partial Spearman ρ (or raw Spearman for the knockout vs. overexpression test) with 95% confidence intervals. The gray band marks the ± 0.05 CI floor: a test passes only if its CI lower bound clears this band in the predicted direction. Same-modality concordance has a larger point estimate ($\rho = 0.25$) than cross-modal ($\rho = 0.19$) but fails because its CI lower bound (0.027) falls inside the floor at $n = 76$. Only cross-modal concordance passes both pre-registered criteria; the seven primary failures delimit what DI measures.

mechanisms—loss-of-function versus gain-of-function—and those mechanisms interact with cellular context independently (Badonyi & Marsh, 2025; Chandrasekaran et al., 2024).

This result constrains the theoretical scope of DI. A gene is context-dependent in a particular way (knockout, overexpression, pharmacological inhibition), and DI values from one perturbation type carry no information about another. A DI atlas must be stratified by perturbation mechanism.

4.3 Proxy explanations fail

Three plausible proxies for context-dependence—expression variance across cell lines, paralog count, and paralog expression breadth—were tested in Batch 2A. All three fail after confound control, and the failures are informative about the structure of the confounds.

Expression variance. Partial Spearman $\rho = 0.060$ (95% CI $[0.038, 0.082]$, $p = 1.1 \times 10^{-7}$, $n = 7{,}802$), controlling for n_{contexts}. The correlation is statistically unambiguous but biologically negligible: $\rho = 0.060$ explains less than 0.4% of variance. The CI lower bound (0.038) does not clear the 0.05 floor. Expression variance captures something genuine yet peripheral—genes with variable expression do produce marginally more variable knockout phenotypes—and the remaining DI variation is driven by something expression variance fails to index. Expression data from DepMap 24Q4 (OmicsExpressionProteinCodingGenesTPMLogp1.csv, 1,673 cell lines, 19,193 genes).

Paralog count. Partial Spearman $\rho = -0.007$ (95% CI $[-0.031, 0.017]$, $p = 0.57$, $n = 6{,}894$), controlling for n_{contexts} and mean expression level. The raw Spearman correlation was $\rho = 0.038$ ($p = 0.002$)—a statistically significant positive association. The partial correlation is zero. The raw association existed because highly expressed genes have more paralogs (gene family expansion correlates with expression level) and independently have higher DI. Controlling for expression removes the confound entirely. This is a textbook example of confound removal: a significant raw association ($p = 0.002$) that evaporates under covariate control. Paralog data from Ensembl BioMart (GRCh38, Ensembl 113).

Paralog expression breadth. Among genes with ≥ 1 expressed paralog ($n = 6{,}882$), partial Spearman $\rho = -0.029$ (95% CI $[-0.053, -0.006]$, $p = 0.015$), controlling for n_{contexts} and mean expression level. The sign is correct (broader paralog compensation predicts lower DI, as hypothesized) and $p = 0.015$ falls below the Bonferroni threshold ($\alpha = 0.0167$). The CI upper bound (-0.006), however, does not clear -0.05. As with expression variance, the CI gate blocks a trivially small effect that would pass on p-value alone. The raw correlation ($\rho = -0.116$, $p \sim 10^{-22}$) is substantially larger, indicating that expression-level confounds inflate the apparent paralog-breadth effect as well.

4.4 Drug selectivity (underpowered)

Partial Spearman $\rho = 0.144$ (95% CI $[-0.088, 0.360]$, $p = 0.22$, $n = 75$), controlling for LINCS n_{contexts}. The primary metric is magnitude-corrected Gini of PRISM kill depth, residualizing raw Gini against mean kill magnitude to remove the mechanical confound between potency and selectivity. The correction reduced the apparent association from $\rho = 0.275$ (raw Gini) to $\rho = 0.144$ (corrected Gini), confirming the magnitude confound is real.

The CI spans $[-0.088, 0.360]$, consistent with anything from no effect to a moderate one. The pre-registration assumed $n \sim 1{,}000$; actual overlap between LINCS drugs and PRISM compounds with $|\text{mean LFC}| \geq 0.01$ is $n = 75$, an order of magnitude smaller. This experiment cannot distinguish a real $\rho = 0.15$ effect from zero.

4.5 Summary: DI is irreducible

The Batch 2A nulls, combined with the Batch 1 essentiality sign-flip and mechanism-specificity result, establish that DI variation across genes and drugs is not explained by existing gene annotations. Essentiality predicts the sign of DI (low) but through a mechanism (functional constraint) that the pre-registration did not anticipate. Expression variance, paralog count, and paralog breadth each contribute less than 0.4% of variance after confound control. Drug selectivity remains unresolved due to insufficient sample size. The picture that emerges is one of specificity: DI captures how a perturbation mechanism interacts with the cell's compensatory network, and this interaction is orthogonal to the gene-level features tested here.

5 Pre-registration integrity

All hypotheses, decision criteria, statistical methods, and analysis code were frozen before DI was computed on new datasets. Batch 1 (concordance, essentiality, knockout vs. overexpression, PRISM) was frozen at commit SHA `0d07a01`; Batch 2A (expression variance, drug selectivity, paralog count, paralog breadth) at commit SHA `a17c125`. The pre-registration documents specify the partial Spearman method, Bonferroni correction levels, CI floor thresholds, power analyses, and interpretation guides for each outcome. Both documents are included in the repository.

5.1 The CI floor

At sample sizes above $n = 5{,}000$, trivially small effects reach statistical significance. The pre-registration introduced a uniform CI lower-bound gate ($|\text{CI bound}| > 0.05$ in the predicted direction) to prevent this. The gate blocked two results (expression variance, paralog breadth) that would have passed on p-value alone, with $\rho = 0.060$ and $\rho = -0.029$ respectively. In both cases, the blocked effects explain less than 0.4% of variance and lack biological plausibility as primary explanations for DI variation. The CI gate performed as designed.

5.2 Deviations

Seven deviations from the pre-registration occurred during analysis. All are data-wrangling corrections or sample-size discrepancies; none alter the statistical method, decision criteria, or interpretation framework.

DepMap version. All DepMap analyses use the pre-registered 24Q4 release (figshare article 27993248): Chronos dependency scores (file 51064667; 1,178 cell lines, 17,916 genes) for the essentiality test, expression (file 51065489; 1,673 cell lines, 19,193 genes) for expression variance and paralog tests. Initial runs inadvertently used 22Q2 Chronos (essentiality) and 24Q2 expression (expression variance, paralogs). Re-running on the correct 24Q4 release changed no verdicts: the essentiality test moved from $\rho = -0.330$ to -0.332, expression variance from 0.064 to 0.060, and paralog count/breadth were identical to three decimal places.

Same-modality concordance matching method. The pre-registration specified InChIKey matching for drug concordance. An initial run erroneously used case-insensitive drug-name matching ($n = 114$, $\rho = -0.003$), which introduced approximately 38 spurious pairs from salt forms, stereoisomers, and name collisions. Reverting to InChIKey matching via PubChem CID cross-reference ($n = 76$, $\rho = 0.254$) moved toward the pre-registered method, resolving the spurious null. The name-matched result is reported in Section 3 as a documented deviation; the InChIKey result is the pre-registered analysis.

Other deviations. (i) Cross-modal concordance yielded $n = 378$ instead of the estimated $n = 330$, because `pert_iname` matching recovered more valid InChIKey pairs than anticipated. (ii) `pert_iname` was used instead of `pert_id` for LINCS metadata lookup; `pert_id` matching was a data-wrangling error yielding only 8 compounds. (iii) LINCS and Tahoe contexts are cell lines; JUMP-CP CRISPR and ORF contexts are plate replicates. This asymmetry in context definitions is inherent to the datasets and does not affect within-dataset DI computation, but cross-dataset comparisons should be interpreted with this caveat. (iv) Drug selectivity yielded $n = 75$ instead of the estimated $n \sim 1{,}000$ due to the small LINCS-PRISM overlap after viability filtering. (v) Nine genes were dropped from the paralog tests due to missing expression values, leaving $n = 6{,}894$ (paralog count) and $n = 6{,}882$ (paralog breadth).

6 Discussion

6.1 DI as an irreducible axis

Nine pre-registered experiments produce a consistent characterization of direction instability. DI is concordant across measurement modalities, confirming that it reflects a biological property of perturbation-context interactions. It is mechanism-specific: knockout and overexpression of the same gene produce unrelated DI profiles, so DI characterizes how a perturbation mechanism interacts with cellular context, not a stable property of the target gene. Functional constraint suppresses DI: essential genes produce stereotyped disruption, yielding low DI. And three plausible proxy explanations—expression variance, paralog count, paralog breadth—each explain less than 0.4% of DI variance after confound control.

This combination of positive and negative results defines DI as an irreducible axis of perturbation characterization. It captures something about how perturbation mechanisms cascade through cell-type-specific compensatory networks, and that something is orthogonal to the gene-level annotations tested here. The irreducibility is itself a characterization of the atlas: users should expect DI to provide information beyond what is available from essentiality screens, expression atlases, or paralog databases.

6.2 The resource

The atlas provides magnitude-corrected DI values with bootstrap confidence intervals for 8,949 LINCS drugs, 379 Tahoe drugs, 7,948 CRISPR knockouts, and 12,590 ORF overexpressions. The companion JUMP-CP compound DI values (25,254 compounds) are available from a prior study (Tower, 2026). Together, these cover the perturbation-biology datasets most widely used for drug mechanism prediction, gene function annotation, and virtual screening benchmarks.

The atlas is designed for downstream integration. Each row provides the columns needed for a secondary analysis: corrected DI, raw DI, CI bounds, mean signature norm, magnitude CV, and number of contexts. Pre-registration documents, analysis scripts, and result JSON files are released alongside the data.

6.3 Limitations

Several limitations constrain the scope of these results.

The PRISM viability analysis ($\rho = 0.359$, $n = 75$) is exploratory: uncorrected p-value, no covariate adjustment, small sample. It provides corroboration, not evidence, and should carry weight only alongside the primary tests.

The same-modality concordance test (LINCS-Tahoe, $n = 76$) was underpowered by design: the LINCS-Tahoe compound overlap is structurally small. The point estimate ($\rho = 0.254$) is consistent with genuine concordance, but the wide CI ($[0.027, 0.456]$) precludes a firm conclusion.

Context definitions vary across datasets. LINCS and Tahoe use cell lines as contexts; JUMP-CP CRISPR and ORF use plate replicates. Plate replicates sample technical variability within one cell line, while cell lines sample biological variability across cell types. DI computed over plate replicates measures reproducibility of the perturbation within a fixed biological context, whereas DI over cell lines measures context-dependence across biological contexts. Cross-dataset comparisons (both concordance tests) match on context type (both datasets use cell lines), but the CRISPR/ORF DI values used in the essentiality, mechanism-specificity, and proxy tests index technical variability rather than biological context-dependence. Extending the atlas to single-cell CRISPR screens with cell-line diversity (Tahoe Consortium, 2025) would resolve this asymmetry.

The essentiality sign-flip interpretation (functional constraint suppresses DI) is post-hoc. The pre-registration predicted the opposite sign. The mechanistic explanation is plausible and connects to established results in functional genomics (Funk et al., 2022; Ramezani et al., 2025); it was, however, generated after observing the data. A concrete falsifiable prediction follows from this interpretation: among non-essential genes, tissue-restricted expression should positively predict DI (Section 4). Prospective testing of this prediction would convert a post-hoc rationalization into a validated mechanism.

The drug selectivity test is underpowered at $n = 75$ and cannot distinguish a moderate effect from zero. Both underpowered tests (same-modality concordance and drug selectivity) are data-limited: the small n reflects the structural size of the dataset intersection (LINCS-Tahoe compound overlap and LINCS-PRISM viability overlap, respectively), not a design choice. We report them for completeness and for the atlas resource record, not as adjudicated hypotheses. Larger drug-viability datasets or compound selectivity metrics from dose-response screens would provide more informative tests.

6.4 Future directions

Three extensions follow naturally from the current results. First, Batch 2B will test whether DI correlates with Shesha geometric stability (Raju, 2026a,b), a concurrent and independently developed measure of perturbation coherence computed on single-cell data. Second, Gene Ontology enrichment analysis on the directional divergence gene sets from the mechanism-specificity test (1,331 knockout-stable / overexpression-unstable genes; 289 with the opposite pattern) may reveal functional categories associated with mechanism-specific context-dependence. Third, expanding the Tahoe drug-InChIKey annotation would increase the same-modality concordance sample size and provide a properly powered same-modality concordance test.

More broadly, the irreducibility of DI to existing gene annotations suggests that it indexes an aspect of gene function—the structure of the compensatory network a perturbation engages—that current annotation systems do not capture. Whether this structure can be predicted from protein interaction networks, pathway topology, or genomic context remains an open question.

7 Conclusion

Direction instability is concordant across measurement modalities, mechanism-specific, shaped by functional constraint, and orthogonal to expression variance, paralog count, and paralog breadth. This combination of one positive and seven negative pre-registered results defines DI as an irreducible axis of perturbation characterization. The atlas provides magnitude-corrected DI values with bootstrap confidence intervals for 8,949 drugs, 379 single-cell-profiled drugs, 7,948 CRISPR knockouts, and 12,590 ORF overexpressions, ready for integration with drug-mechanism prediction, gene-function annotation, and screening quality control.

Availability of source code and requirements

All analysis scripts are written in Python 3 with standard scientific libraries (NumPy, SciPy, pandas, scikit-learn) and are available at `https://github.com/elliottower/direction-instability-atlas` under the MIT License.

Data availability

All atlas data files, experiment result files, pre-registration documents, and analysis code are deposited on Zenodo under a CC-BY-4.0 license. Pre-registration commits can be verified against the repository history at the SHA values reported in the text. LINCS L1000 compound DI values (8,949 drugs) and JUMP-CP compound DI values (25,254 compounds) were previously computed and are available from the companion study (DOI: 10.5281/zenodo.21213700). Source datasets are available from their original repositories: LINCS L1000 (GEO: GSE92742), Tahoe-100M (HuggingFace: tahoebio/Tahoe-100M), JUMP Cell Painting (Cell Painting Gallery, cpg0016), DepMap 24Q4 (figshare: 27993248), PRISM (figshare: Repurposing Public 24Q2), Ensembl paralogs (BioMart, GRCh38).

Declarations

Ethics approval and consent to participate. This study uses exclusively publicly available, summary-level perturbation screening data. No individual-level data, human subjects, or animal subjects were involved. No ethical approval was required.

Competing interests. The author declares no competing interests.

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Authors' contributions. E.T. conceived the study, designed and pre-registered the hypotheses, collected and processed data, wrote the analysis code, performed all analyses, and wrote the manuscript.

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A Pre-registration protocol summary

All hypotheses, decision criteria, statistical methods, and analysis code were frozen before DI was computed on new datasets. The two pre-registration documents are included in the Zenodo deposit.

Table 3: Pre-registration batches.

Batch	Commit SHA	Experiments
Batch 1	`0d07a01`	A1 (LINCS–Tahoe concordance), A2 (LINCS–JUMP concordance), C (essentiality), D (KO vs. OE), PRISM (exploratory)
Batch 2A	`a17c125`	E2 (expression variance), E3 (drug selectivity), E5a (paralog count), E5b (paralog breadth)

Two Batch 1 experiments were dropped before analysis: A3 (Tahoe–JUMP concordance; zero compound overlap between clinical drugs and screening library) and B (MOA stratification in Tahoe; structurally underpowered at $n = 7$ vs. $n = 17$). Batch 2A labels E2/E3/E5 are non-consecutive because the pre-registration proposed three specific explanatory hypotheses, not a sequential series.

What did not change across batches. The DI computation (Equation 1), the magnitude correction (Equation 2), the partial Spearman method (Pearson-residualize then rank), the CI floor gate (± 0.05), and the Bonferroni correction framework were constant across both batches. Batch 2A added three new hypotheses with their own $\alpha = 0.0167$ (Bonferroni for three tests).

B Knockout vs. overexpression permutation null

The mandatory permutation null for the knockout vs. overexpression DI correlation used 1,000 label shuffles preserving plate structure within each dataset. The null distribution had mean = 0.0003, SD = 0.014, and 95th percentile = 0.023. The observed $\rho = 0.011$ falls at permutation $p = 0.215$, well within the null. At $n = 5{,}220$, the study had power to detect effects as small as $\rho = 0.04$ (SE ≈ 0.014), so the null result is well-powered.

C Same-modality concordance matching method comparison

The pre-registration specified InChIKey matching for drug concordance. An initial run erroneously used case-insensitive drug-name matching, which inflated the sample size and introduced spurious pairs.

Reverting to InChIKey matching resolved the spurious null. The name-matched result ($\rho = -0.003$) reflects contamination by chemically distinct compounds sharing common drug names, not absence of concordance.

D Magnitude correction details

The OLS magnitude correction (Equation 2) is applied independently per dataset. For each dataset, raw DI is regressed on mean signature L_2 norm. The residuals plus the intercept yield $\mathrm{DI}_{\mathrm{corrected}}$.

Table 4: Same-modality concordance results by matching method. InChIKey matching is the pre-registered analysis.

Method	Match type	n	Partial ρ	Note
Drug-name (erroneous)	Case-insensitive	114	−0.003	∼38 spurious pairs from salt forms, stereoisomers
InChIKey (pre-registered)	Connectivity prefix	76	0.254	Via PubChem CID cross-reference

Table 5: Magnitude correction parameters per dataset. Negative slopes indicate that weaker perturbations (lower norm) have higher raw DI, as expected from the noise-inflation mechanism.

Dataset	n	OLS slope	R^2
Tahoe-100M	379	+0.00013	0.002
JUMP-CP CRISPR	7,946	−0.0097	0.081
JUMP-CP ORF	12,590	−0.0126	0.182

LINCS L1000 and JUMP-CP compound DI values were corrected in the companion study (Tower, 2026) and are distributed with pre-corrected values. The near-zero Tahoe slope reflects Tahoe-100M's high signal-to-noise ratio (deeply sequenced single-cell data); the correction has negligible effect on this dataset.